$$f(x) = \sqrt[3]{x^2-1}\cdot e^{-x}$$

D: $\mathbb{R}$

$$\lim_{x\to-\infty} \sqrt[3]{x^2-1}\, e^{-x} = +\infty$$

$$\lim_{x\to+\infty} \sqrt[3]{x^2-1}\, e^{-x} = \lim_{x\to+\infty} \frac{\sqrt[3]{x^2-1}}{e^x} = 0$$

$y = 0$ AS. OR. $x \to +\infty$

$$m = \lim_{x\to-\infty} \frac{\sqrt[3]{x^2-1}}{x} e^{-x} = \lim_{x\to-\infty} \frac{x^{2/3}}{x} e^{-x}$$

$$= \lim_{x\to-\infty} x^{-1/3} e^{-x} = \lim_{x\to-\infty} \frac{e^{-x}}{x^{1/3}} = -\infty$$

non c'è AS OBL.

Segno e zeri

$f(x) \geq 0 \quad \sqrt[3]{x^2-1}\, e^{-x} \geq 0$

$e^{-x} > 0 \ \forall x \quad \sqrt[3]{x^2-1} \geq 0$

$x^2 - 1 \geq 0 \quad x \leq -1 \vee x \geq 1$

$f(x) > 0 \quad x < -1 \vee x > 1$

$f(x) = 0 \quad x = -1 \quad x = 1$

$f(x) < 0 \quad -1 < x < 1$

$$f(x) = (x^2-1)^{1/3} e^{-x}$$

$$f'(x) = \frac{1}{3}(x^2-1)^{-2/3} \cdot 2x \cdot e^{-x} - e^{-x}(x^2-1)^{1/3}$$

$$= e^{-x}\left(\frac{1}{3(x^2-1)^{2/3}} \cdot 2x - (x^2-1)^{1/3}\right)$$

$$= e^{-x}\left(\frac{2x - 3(x^2-1)}{3(x^2-1)^{2/3}}\right)$$

D': $x^2 - 1 \neq 0 \quad x \neq \pm 1$

Studio $x = -1$

$$\lim_{x \to -1^{\pm}} e^{-x} \frac{(-3x^2+2x+3)}{3\sqrt[3]{(x^2-1)^2}} = -\infty$$

FLESSO A TG. VERT

Studio $x = 1$

$$\lim_{x \to 1^{\pm}} \frac{e^{-x}(-3x^2+2x+3)}{3\sqrt[3]{(x^2-1)^2}} = +\infty$$

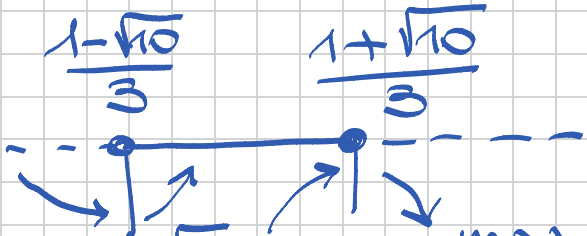
FLESSO A TG. VERT

$f'(x) \geq 0 \qquad \frac{e^{-x}(2x-3x^2+3)}{\sqrt[3]{(x^2-1)^2}} \geq 0$

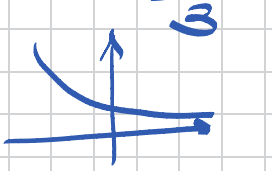
$e^{-x} > 0 \quad \forall x \qquad 2x-3x^2+3 \geq 0$

$3x^2-2x-3 \leq 0 \qquad x_{1,2} = \frac{1 \pm \sqrt{10}}{3}$

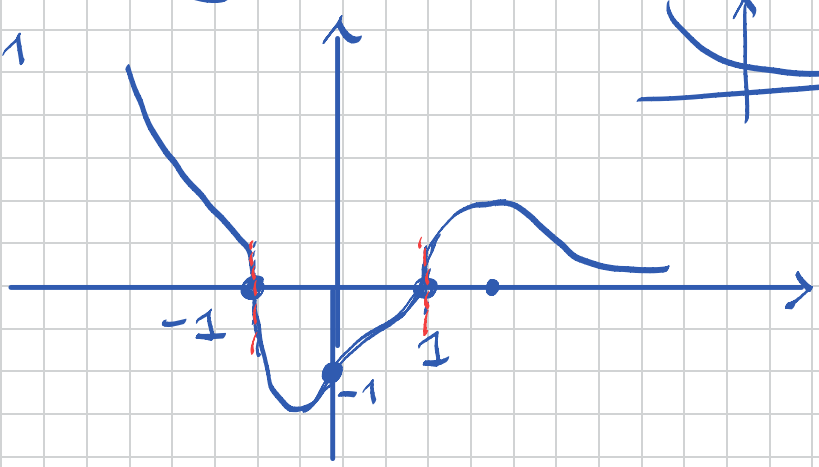
$\frac{1-\sqrt{10}}{3} \leq x \leq \frac{1+\sqrt{10}}{3} \qquad D > 0 \ \forall x \neq \pm 1$



min $x = \frac{1-\sqrt{10}}{3}$ max $x = \frac{1+\sqrt{10}}{3}$



$f(0) = -1$



$f(x) = e^{\frac{1}{x}}(x+3)$

D: $x \neq 0$

$\lim_{x \to \pm\infty} e^{\frac{1}{x}}(x+3) = \pm\infty$

$\lim_{x \to 0^-} e^{\frac{1}{x}}(x+3) = 0$

$\lim_{x \to 0^+} e^{\frac{1}{x}}(x+3) = +\infty$ $x=0$ A.V. Destro

ASINTOTO OBLIQUO

$m = \lim_{x \to \pm\infty} \frac{e^{\frac{1}{x}}(x+3)}{x} = 1$

$q = \lim_{x \to \pm\infty} e^{\frac{1}{x}}(x+3) - x$

$\lim_{x \to \pm\infty} x\left(e^{\frac{1}{x}} - 1\right) + 3e^{\frac{1}{x}}$

$\lim_{x \to \pm\infty} x \cdot \frac{1}{x} + 3e^{\frac{1}{x}} = 4$

$\Rightarrow y = x+4$

Segno e zeri $f(x) \geq 0$

$e^{\frac{1}{x}}(x+3) \geq 0$ $e^{\frac{1}{x}} > 0 \ \forall x \in D$

$x \geq -3$ $f(x) \geq 0$ $x \geq -3$

$f(x) = 0$ $x = -3$

$$f'(x) = e^{\frac{1}{x}}\left(-\frac{1}{x^2}\right)(x+3) + e^{\frac{1}{x}}$$

$$= e^{\frac{1}{x}}\left(-\frac{1}{x} - \frac{3}{x^2} + 1\right) =$$

$$= e^{\frac{1}{x}}\left(\frac{-x-3+x^2}{x^2}\right)$$

D': $x \neq 0$

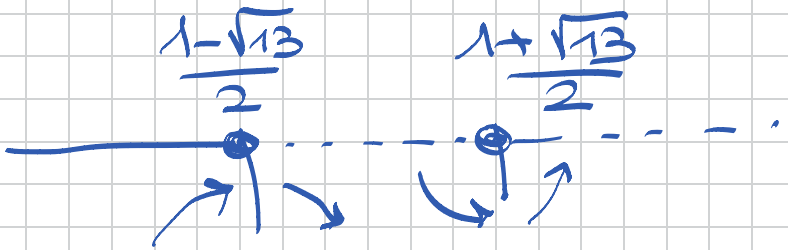
$f'(x) \geq 0 \qquad e^{\frac{1}{x}}\left(\frac{x^2-x-3}{x^2}\right) \geq 0$

$e^{\frac{1}{x}} > 0 \quad \forall x \in D$

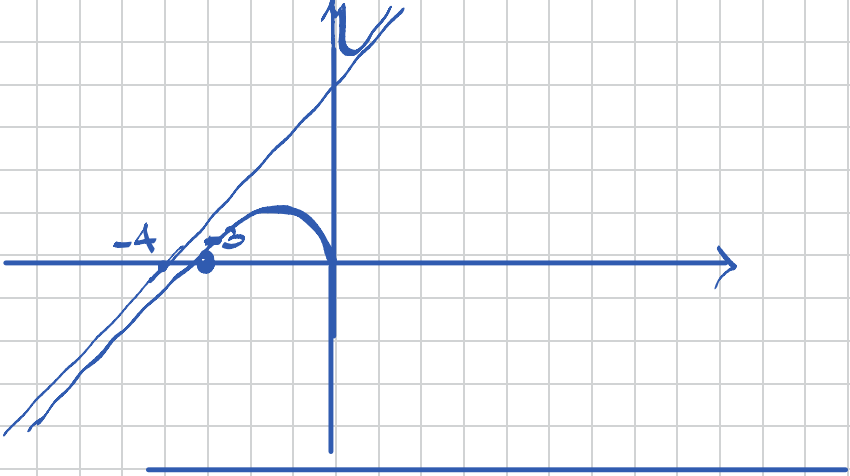
$x^2 - x - 3 \geq 0 \qquad x_{1,2} = \frac{1 \pm \sqrt{13}}{2}$

$x^2 > 0 \quad \forall x \neq 0$

$x \leq \frac{1-\sqrt{13}}{2} \vee x \geq \frac{1+\sqrt{13}}{2}$



max $x = \frac{1-\sqrt{13}}{2}$ min $x = \frac{1+\sqrt{13}}{2}$



$$f(x) = \log(1+x^2) + \arctan\frac{1}{x}$$

D: $1+x^2 > 0 \quad \forall x$

$x \neq 0$

$$\lim_{x\to\pm\infty} \log(1+x^2) + \arctan\frac{1}{x} = +\infty$$

$$\lim_{x\to 0^-} \log(1+x^2) + \arctan\frac{1}{x} = -\frac{\pi}{2}$$

$$\lim_{x\to 0^+} \log(1+x^2) + \arctan\frac{1}{x} = \frac{\pi}{2}$$

$$m = \lim_{x\to\pm\infty} \frac{\log(1+x^2) + \arctan\frac{1}{x}}{x} = 0$$

No asintoto oblicuo

$$f'(x) = \frac{1}{1+x^2}\cdot 2x + \frac{1}{1+\frac{1}{x^2}}\cdot\left(-\frac{1}{x^2}\right)$$

$$f'(x) = \frac{2x}{1+x^2} + \frac{1}{\frac{x^2+1}{x^2}} \cdot \left(-\frac{1}{x^2}\right)$$

$$= \frac{2x}{1+x^2} + \frac{x^2}{x^2+1}\left(-\frac{1}{x^2}\right) = \frac{2x-1}{x^2+1}$$

$D' : \mathbb{R}$

$f'(x) \geq 0 \quad \frac{2x-1}{x^2+1} \geq 0 \qquad D>0 \ \forall x$

$2x-1 \geq 0 \quad x \geq \frac{1}{2}$



Min $x = \frac{1}{2}$

$f(1/2) = \underbrace{\log\left(1+\frac{1}{4}\right)}_{+} + \underbrace{\arctan 2}_{+} > 0$

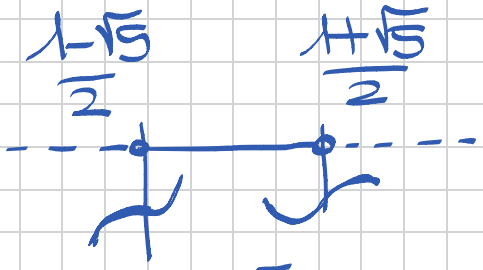
$$f''(x) = \frac{2(x^2+1) - 2x(2x-1)}{(x^2+1)^2} = \frac{-2x^2+2x+2}{(x^2+1)^2}$$

$$f''(x) \geq 0 \quad \frac{-2x^2+2x+2}{(x^2+1)^2} \geq 0$$

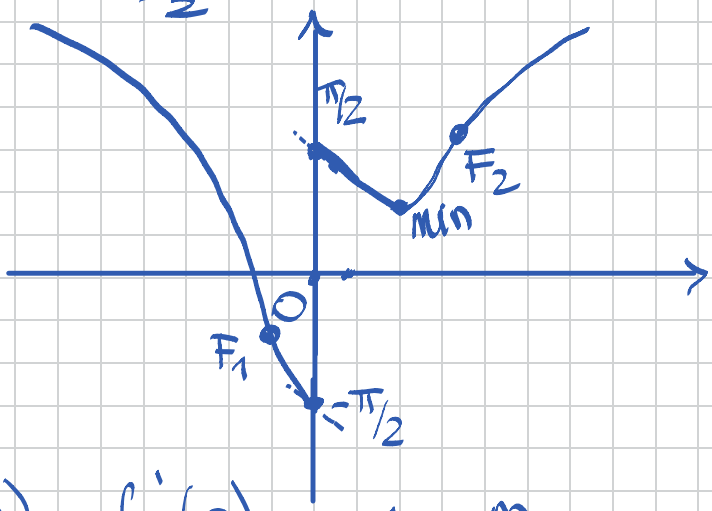
$-x^2+x+1 \geq 0 \qquad D>0 \ \forall x$

$x^2-x-1 \leq 0 \qquad x_{1,2} = \frac{1 \pm \sqrt{5}}{2}$

$\frac{1-\sqrt{5}}{2} \leq x \leq \frac{1+\sqrt{5}}{2}$



F_1 $x = \frac{1-\sqrt{5}}{2}$ F_2: $x = \frac{1+\sqrt{5}}{2}$

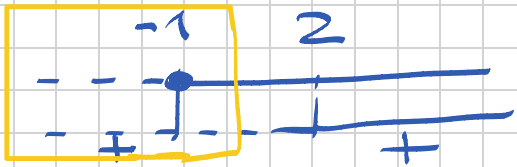


$f'_-(0) = f'_+(0) = -1 = m_{tg}$

$$f(x) = \begin{cases} \sqrt{\frac{x+1}{x-2}} & x \le -1 \\ \frac{2x+2}{x-3} & x > -1 \end{cases}$$

$$f_1(x) = \sqrt{\frac{x+1}{x-2}} \qquad \frac{x+1}{x-2} \ge 0$$

$x \ge -1$
$x > 2$



$f_2(x) = \frac{2x+2}{x-3}$ $x \neq 3$ D: $x > -1$, $x \neq 3$

$\lim_{x \to -\infty} \sqrt{\frac{x+1}{x-2}} = 1$ $y = 1$ AS. OR.

$\lim_{x \to 3^{\pm}} \frac{2x+2}{x-3} = \pm\infty$ $x = 3$ AV.

$\lim_{x \to +\infty} \frac{2x+2}{x-3} = 2$ $y = 2$ AS. OR.

Studio $x = -1$

$\lim_{x \to -1^+} \frac{2x+2}{x-3} = 0$ $f(-1) = 0$

$\lim_{x \to -1^-} \sqrt{\frac{x+1}{x-2}} = 0 = f(-1)$

in $x = -1$ $f(x)$ continua

Segno e zeri:

$f_1(x) \geq 0$ $\sqrt{\frac{x+1}{x-2}} > 0$ $\forall x \in D$

$f_1(x) = 0$ $x = -1$

$f_2(x) \geq 0$ $\frac{2x+2}{x-3} \geq 0$ $N \geq 0$ $x \geq -1$ $D > 0$ $x > 3$



$f_2(x) > 0$ $x > 3$

$f_2(x) \leq 0$ $-1 \leq x < 3$

$$f_1 = \sqrt{\frac{x+1}{x-2}}$$

$$f_1' = \frac{1}{2}\sqrt{\frac{x-2}{x+1}} \cdot \left(\frac{x-2-(x+1)}{(x-2)^2}\right)$$

$$= \frac{1}{2}\sqrt{\frac{x-2}{x+1}} \cdot \left(\frac{-3}{(x-2)^2}\right)$$

D': $\frac{x-2}{x+1} \geq 0 \quad x \neq 2 \quad \boxed{x < -1}$

$\Rightarrow$ devo studiare $x = -1$

$$f_2 = \frac{2x+2}{x-3} \qquad f_2' = \frac{2(x-3)-(2x+2)}{(x-3)^2}$$

$$= \frac{-8}{(x-3)^2} \qquad D': x \neq 3$$

$$f'(x) = \begin{cases} -\frac{3}{2}\sqrt{\frac{x-2}{x+1}} \cdot \frac{1}{(x-2)^2} & x < -1 \\ \frac{-8}{(x-3)^2} & x > -1 \quad x \neq 3 \end{cases}$$

Studio $x = -1$

$\lim_{x \to -1^-} -\frac{3}{2}\sqrt{\frac{x-2}{x+1}} \cdot \frac{1}{(x-2)^2} = -\infty$ TANG. V.

$\lim_{x \to -1^+} \frac{-8}{(x-3)^2} = -\frac{1}{2}$ TANG. OBLIQUA

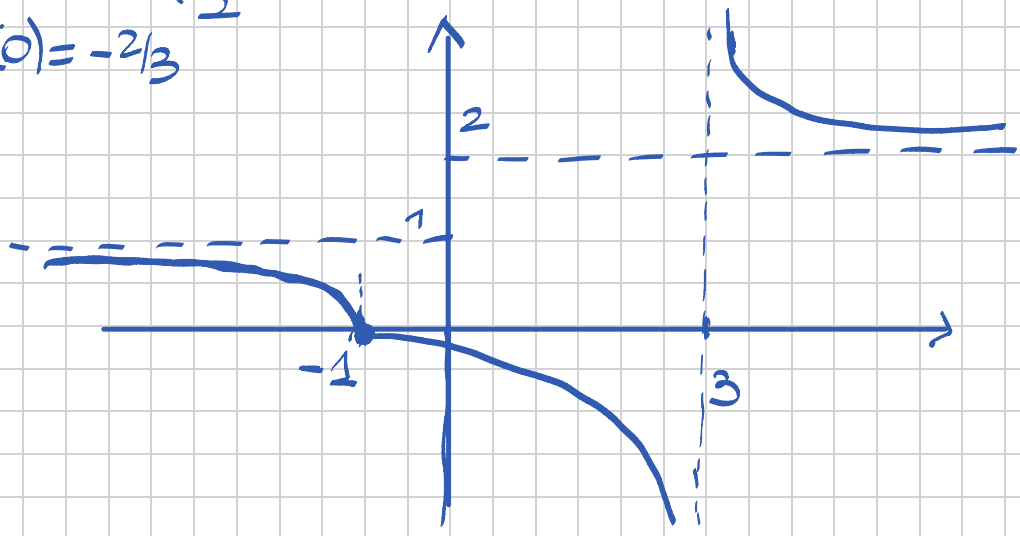
$f_1' \geq 0 \quad -\frac{3}{2}\sqrt{\frac{x-2}{x+1}} \cdot \frac{1}{(x-2)^2} \geq 0$

$\Rightarrow f_1' < 0 \ \forall x \in D \Rightarrow f_1(x)$ è strettam. decrescente in D

$f_2' \geq 0 \quad \frac{-8}{(x-3)^2} \geq 0$ mai

$\Rightarrow f_2$ è strettam. decresc. in D

$f(0) = -2/3$



$$\lim_{n\to+\infty} \frac{1-\cos^2\left(\sin\left[\frac{3}{n}\right]\right)}{\sqrt[3]{n^3+n}-n}$$

$$\lim_{n\to+\infty} \frac{\left(1-\cos\left(\sin\left(\frac{3}{n}\right)\right)\right)\left(1+\cos\left(\sin\left(\frac{3}{n}\right)\right)\right)}{\sqrt[3]{n^3\left(1+\frac{1}{n^2}\right)}-n}$$

$$\lim_{n\to+\infty} \frac{\frac{1}{2}\left(\sin\left(\frac{3}{n}\right)\right)^2\left(1+\cos\left(\sin\left(\frac{3}{n}\right)\right)\right)}{n\left(1+\frac{1}{n^2}\right)^{1/3}-n}$$

$$\lim_{n\to+\infty} \frac{\frac{1}{2}\left(\frac{3}{n}\right)^2\left(1+\cos\left(\sin\left(\frac{3}{n}\right)\right)\right)}{n\left[\left(1+\frac{1}{n^2}\right)^{1/3}-1\right]}$$

$$\lim_{n\to+\infty} \frac{\frac{9}{2n^2}\cdot 2}{n\cdot\frac{1}{3}\cdot\left(\frac{1}{n^2}\right)} = \lim_{n\to+\infty} \frac{9}{n^2}\cdot 3n = 0$$

$$\lim_{n\to+\infty} \frac{\left(3+\frac{1}{n}\right)^n - n^3}{3^n + \log(1+n^n)}$$

$$\lim_{n\to+\infty} \frac{\left[3\left(1+\frac{1}{3n}\right)\right]^n - n^3}{3^n + \log(1+n^n)}$$

$$\lim_{n\to+\infty} \frac{3^n\left(1+\frac{1}{3n}\right)^n - n^3}{3^n + \log(1+n^n)}$$

$$\lim_{n\to+\infty} \frac{3^n\left[\left(1+\frac{1}{3n}\right)^{3n}\right]^{\frac{1}{3}} - n^3}{3^n + \log(1+n^n)}$$

$$\lim_{n\to+\infty} \frac{3^n \cdot e^{1/3} - n^3}{3^n + \log(1+n^n)} = e^{1/3} = \sqrt[3]{e}$$

Data l'equazione $|z^4|+1-2\bar{z}^2=0$

a) ha 4 soluz. distinte

b) ha solo 2 soluzioni reali

c) ha solo 2 soluzioni puramente immaginarie

d) ha solo una coppia di soluzioni complesse coniugate

e) nessuna delle altre affermazioni è corretta

$z=\rho e^{i\theta}$ $\quad \rho^4+1-2\rho^2 e^{-2i\theta}=0$

$\bar{z}=\rho e^{-i\theta}$ $\quad (\rho^4+1)e^{i0}=2\rho^2 e^{-2i\theta}$

$$\begin{cases} \rho^4+1=2\rho^2 \\ 0=-2\theta+2k\pi \quad k=0,1 \end{cases}$$

$$\begin{cases} \rho^4-2\rho^2+1=0 \\ \theta=k\pi \quad k=0,1 \end{cases} \qquad \begin{cases} (\rho^2-1)^2=0 \\ \theta=k\pi \quad k=0,1 \end{cases}$$

$$\begin{cases} \rho^2=1 \Rightarrow \rho=1 \\ \theta=k\pi \quad k=0,1 \end{cases} \qquad z_1=e^{i0}=1 \qquad z_2=e^{i\pi}=-1$$

$z^4 = z(-4\sqrt{3} + 4i)$

$z = 0$ Soluzione

$z^3 = -4\sqrt{3} + 4i$

$-4\sqrt{3} + 4i = w$ $\quad |w| = 8$

$\tan\theta = \frac{4}{-4\sqrt{3}} = -\frac{\sqrt{3}}{3}$ $\quad \theta = \frac{5}{6}\pi$

$|w| = \sqrt{(-4\sqrt{3})^2 + 4^2}$

$= \sqrt{48+16}$

$w = 8e^{i\frac{5}{6}\pi}$

$z^3 = \rho^3 e^{3i\theta}$ $\quad z^3 = w$

$\rho^3 e^{3i\theta} = 8e^{i\frac{5}{6}\pi}$

$$\begin{cases} \rho^3 = 8 \\ 3\theta = \frac{5}{6}\pi + 2k\pi \end{cases} \qquad \begin{cases} \rho = 2 \\ \theta = \frac{5}{18}\pi + \frac{2}{3}k\pi \end{cases}$$

$k = 0, 1, 2$

z_0 soluz. $z_0 i$ $\quad z_0 e^{i\frac{2}{3}\pi}$